Kidney & UTUC

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 136	Benign Renal Tumours
Ch 137	Malignant Renal Tumours
Ch 138	Urothelial Tumours of the Upper Tract & Ureter
Ch 139	Surgical Management of UTUC
Ch 141	Contemporary Open Surgery of the Kidney
Ch 142	Laparoscopic & Robotic Surgery of the Kidney (incl. donor nephrectomy)
Ch 143	Focal Therapy for Renal Tumours
Ch 144	Treatment of Advanced RCC
★	Putting it together

136 Benign Renal Tumours

1 · The benign small renal mass

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Only **simple cyst** and **fat-containing AML** are imaging diagnoses; size, sex, age unreliable.
- Management follows RCC risk: no follow-up (simple cyst) → SAE (large AML) → excision (solid/complex cystic). Biopsy, SRM algorithm → *Ch 137*.

2 · Renal cysts and the Bosniak classification

- **Simple cyst**: any size, homogeneous, **≤20 Hounsfield units (HU)** → no further imaging. Otherwise → Bosniak.

Class	Features (v2019)	Malignancy	Management (CUA 2023)
I	Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (**3 mm**) enhancing wall/septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI	Likely benign	Active surveillance up to **5 yr**
III	Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall/septum	**~50%**	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	**~90%**	

KEY POINT
Ladder: wall ≤2 → 3 → ≥4 mm → irregular → nodule. Traps: any calcification fits II; I wall may enhance; large hyperdense cysts no longer IIF; I/II growth not malignant.

KEY POINT
Why surveil small III/IV? III has low PPV; cystic RCC indolent (mostly fluid, size predicts poorly; proposed **pT1c**) → very high number needed to treat.

- EAU: Bosniak IV = localised solid mass; biopsy only if solid component and result changes management; ablation of cystic masses unproven.

Diffuse cystic disease

- P2 ADPKD (Adult polycystic kidney disease); Acquired renal cystic disease — *genetics, extrarenal cysts, US criteria, tolvaptan, cyst complications; ACKD prevalence, ADPKD vs ACKD*
- ACKD: cysts from proximal tubule; **80%** by 10 yr of dialysis; RCC risk **≥10×** (multifocal, bilateral, papillary commoner; distinct WHO 2022 entity).

- ADPKD: RCC not commoner but earlier, sarcomatoid, harder to diagnose. PKD1 relatives 15–29: **≥3 cysts**; normal scan excludes only from age 40. Cyst infection: FDG-PET.

3 · Oncocytoma

- **Commonest benign enhancing renal mass**; up to half of benign SRMs <4 cm.
- **Birt–Hogg–Dubé** (folliculin, chr 17): fibrofolliculomas, lung cysts/pneumothorax, oncocytoma, chromophobe RCC (chRCC). Oncocytosis (diffuse bilateral nodules) may harbour chRCC/hybrid tumours.
- **Stellate scar** non-specific; **segmental enhancement inversion** favours oncocytoma.

	Oncocytoma	Chromophobe RCC
CK7	Rare, single-cell; KIT positive	Diffuse positive
Cytogenetics	Normal, −1, −Y, 11q13 rearrangement	Extensive abnormalities

KEY POINT

Why can a core not clear oncocytoma? Sampling error; chRCC has oncocytic areas — a minority of biopsy "oncocytomas" are RCC. EAU: no definitive core diagnosis; CK7 and cytogenetics help.

4 · Angiomyolipoma

- P3 P2 Management of AML; Renal aangiomyolipoma — *PEComa, TSC, mTOR, fat on CT, epithelioid variant, triggers, SAE, NSS, everolimus*
- Multiple tumours clonally independent. Giant AML arises from parenchyma; liposarcoma envelops kidney.

ALGORITHM

Spontaneous retroperitoneal haemorrhage (Wunderlich†) → resuscitate → **SAE** → nephrectomy only if SAE unavailable/unsuitable

Symptomatic (pain, haematuria, barriers to care) → SAE → NSS or RN → ablation in selected cases

Asymptomatic sporadic: >**6 cm**, aneurysm **≥5 mm**, growth >**0.25 cm/yr**, childbearing age/pregnancy → treat; otherwise surveil

TSC: >**3 cm** or growing → everolimus first line; SAE or kidney-sparing resection second

KEY POINT

Why abandon 4 cm? Poorly specific → overtreatment. <4 cm → surveil; observed ≥4 cm → CT angiography for aneurysm. "Barriers to care": surveillance needs access to imaging/emergency care. Pregnancy → treat before conception.

- TSC haemorrhage → embolise then corticosteroids; avoid nephrectomy. EAU (sporadic): mTOR inhibitor only if unfit for SAE/surgery.

5 · Other benign epithelial and mesenchymal tumours

- **Papillary adenoma**: unencapsulated, **≤15 mm**; shares +7, +17, −Y with papillary RCC; cannot metastasise; acceptable in donor kidney.
- **Metanephric adenoma**: women, 50s; polycythaemia. Hypovascular; biopsy unproven → NSS. **WT1, CD57 positive; CK7, AMACR negative**; lacks +7/+17/−Y.

- **MEST and adult cystic nephroma**: one WHO 2022 family; middle-aged women; ER/PR-positive stroma; manage by Bosniak. Paediatric cystic nephroma is distinct (**DICER1**).
- **Leiomyoma**: from renal capsule; desmin, caldesmon positive. Often really lipid-poor AML: **Melan-A** positive only in AML.

Tumour	Key trait
Haemangioma	Sturge–Weber, Klippel–Trénaunay; usually excised
Haemangioblastoma	VHL; mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria
Reninoma (juxtaglomerular cell tumour)	Young women; refractory hypertension, hypokalaemia, high renin; renin granules; excision cures
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · Landmark trial

Trial	Design	Result	Take-home
EXIST-2	Phase III RCT, everolimus vs placebo, TSC/LAM AML	≥50% volume reduction **42% vs 0%**; no bleeding needing embolisation	Approved; mTOR inhibition first line in TSC

137 Malignant Renal Tumours

1 · Overview, epidemiology & aetiology

- Renal cell carcinoma (RCC) = 2–3% of adult cancers; most lethal common urological cancer. M:F **2:1**; peak 55–75 y; **4–6% familial**.
- >60% incidental, ~70% organ-confined; incidence rising faster than mortality → avoid overtreatment.
- Risk factors: **tobacco** (most accepted; RR ~1.4, dose-dependent, falls after quitting), **obesity** ("obesity paradox": higher BMI but lower stage), **hypertension** (the disease, not its drugs); CKD/ESRD; family history 2–4×.
- Origin: clear cell & papillary → **proximal tubule**; chromophobe & collecting duct → distal nephron.
- Children/young adults: RCC 1–6% of paediatric renal tumours (Wilms ~90%), equals Wilms in 2nd decade; commonest = **Xp11/TFE3 translocation RCC**; do well with surgery even if node-positive → formal lymph node dissection (LND) at radical nephrectomy (RN).

2 · Classification & pathology

WHO 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC NOS
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; FH-deficient; SDH-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; **epithelioid angiomyolipoma**; metastases

KEY POINT

Trap: epithelioid AML is malignant; classic AML, papillary adenoma, oncocytoma benign. Papillary type 1/2 split **abandoned** — "type 2" was several diseases (MET-altered, MiT translocation, FH-deficient) with different prognoses → WHO moved to molecular definitions.

Gross features

- **Pseudocapsule** (compressed parenchyma); not infiltrative except collecting duct and sarcomatoid. Cystic degeneration 10–25% (better prognosis); calcification ~10%.
- Locally aggressive in ~20% (perinephric fat, sinus, collecting system); Gerota's usually blocks adjacent organs. Venous involvement ~10% (more than any tumour), up to right atrium. Bilateral 2–4% sporadic; multicentric 10–20% — occult satellites are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtypes

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Hypervascular; clear cytoplasm; CA-IX+; sarcomatoid 5–8%	Most venous invasion; worse localised, better metastatic
Papillary (10–15%)	**Trisomy 7, 17**, loss Y; MET altered 81%	Hypovascular; CK7, AMACR; multicentric up to 40%	Good (5-y CSS 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in BHD	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	Intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; **cup-like CA-IX**	ESRD, VHL; now benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8+; diagnosis of exclusion	Poor; urothelial-like → may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	**Sickle cell trait**	Young Black patients; infiltrative	Dismal, median 8–13 mo; modest platinum response
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss, 2SC+	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells + papillae; FISH confirms	40% of paediatric RCC; t(X;1) late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test ALK before "unclassified"	Dramatic responses to ALK inhibitors in some
Unclassified (1–5%)	—	Shrinking catch-all	Poor

- **Indolent:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC, clear cell papillary. **Lethal:** medullary, FH-deficient, collecting duct, unclassified, sarcomatoid.
- Chromophobe vs oncocytoma hard → WHO 2022 "oncocytic renal neoplasm of low malignant potential". Papillary adenoma ≤15 mm benign.
- **Sarcomatoid** (1–5%): dedifferentiation of any subtype; median survival <1 y. Sarcomatoid/rhabdoid = automatic grade 4; report in every RCC.

> **KEY POINT**
> Localised clear cell does worse than papillary/chromophobe (intrinsically aggressive), yet metastatic clear cell does better: VHL-driven angiogenesis and immunogenicity → responds to TKIs and immune checkpoint inhibitors (ICIs).

WHO/ISUP grading — replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100

Grade	Criterion
3	Nucleoli conspicuous, eosinophilic at **×100**
4	Extreme pleomorphism, giant cells, **rhabdoid/sarcomatoid**

Nuclear grade is an independent prognostic factor.

3 · Molecular biology & hereditary syndromes

- P2 P1 VHL gene and von Hippel–Lindau disease; Familial malignancies in urology — *VHL–HIF pathway, two-hit, VHL lesions, genotype–phenotype, VHL screening, 3-cm rule, belzutifan; VHL/HPRC/HLRCC/BHD/TSC genes and features*

> **ALGORITHM**
> Normoxia: VHL + elongin B/C + cullin-2 + RBX1 = E3 ligase → HIF-1α/2α degraded
> VHL loss or hypoxia → **HIF-2α** + ARNT → ↑ VEGF, PDGF, TGF-α, GLUT1, EPO, CA-IX → hypervascular clear cell RCC

Drug targets: VEGF ligand (bevacizumab) · VEGFR TKIs (sunitinib, pazopanib, axitinib, cabozantinib) · mTOR (temsirolimus, everolimus) · HIF-2α (**belzutifan**, FDA 2021 for VHL, ~50% response) · CTLA-4 (ipilimumab), PD-1 (nivolumab, pembrolizumab), PD-L1 (atezolizumab, avelumab).

- Multidrug-resistance pumps → chemotherapy response 5–7%; checkpoint blockade now mainstay *→ Ch 144*.
- Poor prognosis in all subtypes: **CDKN2A** (9p) loss, CpG hypermethylation, Th2 signature. BAP1 mutation (up to 14% sporadic clear cell) → high grade, rapid growth, unsuitable for active surveillance (AS); test germline BAP1 in early-onset multifocal VHL-negative clear cell RCC.

Syndrome (beyond topics)	Gene (chr)	Renal tumour	Other / management
HLRCC	FH (1q42–43)	Aggressive, often **solitary**	Uterine leiomyosarcoma; **prompt excision at any size, wide margins**; PN only if wide margin feasible; penetrance 15–20%; FDG-PET finds small metastases
SDH-RCC	SDHA/B/C/D, SDHAF2	Onset ≤40 y	Paraganglioma/phaeo, GIST, thyroid; excision, PN preferred
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast (50%), thyroid; KILLIN if PTEN-negative; CT/MRI, 3-cm rule
Tuberous sclerosis	TSC1/TSC2	AMLs; clear cell 1–3%	AML: surveil <3, everolimus 3–5, embolise/excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma; nephron-sparing excision
MiTF-associated	MiTF p.E318K	RCC >5× (activates HIF-1α)	Melanoma; management undefined

- Hereditary papillary is indolent → avoid CKD from repeated surgery.

4 · Clinical presentation

- **Classic triad** (flank pain, gross haematuria, mass) rare = advanced ("too late triad"). Clues: weight loss, bone pain, cervical nodes, **non-reducing or right-sided varicocele**, leg oedema (IVC obstruction).
- Spontaneous perirenal haematoma: >50% tumour → repeat CT.

Paraneoplastic syndromes (10–20%)

Syndrome	Frequency / mechanism	Key point
Raised ESR	Commonest	Advanced disease marker
Hypercalcaemia	Up to 13%; PTHrP or osteolytic metastases	**Only one manageable medically**: hydration → furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy ± metastasectomy; osteolytic type harder (bisphosphonates, radiotherapy)
Hypertension	Renin, artery compression, AV fistula	—
Polycythaemia	Erythropoietin	—
Stauffer syndrome	3–20%; IL-6	Non-metastatic liver dysfunction (↑ ALP almost always); exclude liver metastases; normalises after nephrectomy in 60–70%; persistence = viable tumour, poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders, cerebellar ataxia

- Rule: paraneoplastic syndromes resolve only with excision or systemic therapy (except medically managed hypercalcaemia); persistence after nephrectomy = residual disease.

5 · Diagnosis

- P4 Radiology for solid renal masses — *CT phases and HU thresholds, MRI, CEUS, sestamibi, girentuximab, FDG-PET, biopsy yield, venous thrombus CT sensitivity, R.E.N.A.L./PADUA*
- Strongest malignancy predictors: **male sex (~3×)** and **size** (benign 20–30% of ≤4 cm, ~40% of ≤2 cm; ≥7 cm few benign, half high grade). Nomograms barely beat a coin toss (concordance 0.55–0.65). cT1: ~20% benign · ~60% indolent · ~20% potentially aggressive.
- **Multiphase CT** = single most important test (enhancement **>15–20 HU**; fat **<−20 HU** = AML). MRI best for IVC thrombus and organ invasion; macrocyclic gadolinium safe even on dialysis. Iodinated contrast with eGFR <45 → periprocedural hydration.
- Infiltrative pattern → lymphoma, urothelial, sarcomatoid, collecting duct, medullary, XGP.

ALGORITHM

Simple cyst → observe

Complex/solid: eGFR >45 → CT or MRI with/without contrast · eGFR 30–45 → hydrate, same · eGFR <30 → non-contrast CT or gadolinium MRI · contrast allergy → USG/contrast USG or MRI

Fat → AML · complex cyst → Bosniak · solid enhancing → management options ± renal mass biopsy

- **Bosniak** → *Ch 136*: malignancy IIF 3–10% · III ~50% · IV 75–90%. Key discriminator: unequivocal **measurable enhancement**. Avoid ablation and biopsy of cystic masses (poor yield, spillage); AS acceptable for III/IV.
- **Renal mass biopsy (RMB)**: when result may change management, or lymphoma, metastasis, infection suspected; not for patients managed the same regardless. **Multiple cores over FNA**. Haematoma ~5%, transfusion <0.5%; specificity 94%; non-malignant result ≠ benign. Biopsy before ablation.
- **Screening:** none in population or ADPKD. ESRD (acquired cystic disease 80%; RCC 1–2% of these): USG/CT from 3rd year of dialysis if fit; continue after transplant. Genetic counselling: RCC at **≤46 y**, multifocal/bilateral, suggestive history or pathology.

6 · Staging & prognosis

TNM (AJCC 8th)	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%
T3a	Renal vein/segmental branches, **pelvicalyceal system**, perirenal/sinus fat, within Gerota's	45–75%
T3b	IVC below diaphragm	40–55%
T3c	IVC above diaphragm or **caval wall invasion**	20–45%
T4	Beyond Gerota's, incl. **contiguous ipsilateral adrenal**	0–35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0–30%

Stage I T1N0 · II T2N0 · III T1–2N1 or T3 any N · IV T4 or M1. Changes: 6th split T1 at 4 cm; 7th split T2 at 10 cm, contiguous adrenal → T4, renal vein thrombus T3b → T3a, N1/N2 merged; 8th pelvicalyceal invasion → T3a.

> **KEY POINT**
> TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus supradiaphragmatic thrombus) are lost — record all; use integrated nomograms.

- **Work-up:** abdominal CT + chest X-ray (CT if symptoms or high risk); bone scan only with ↑ ALP, bone pain or poor PS; no PET. Small nodes often reactive — don't preclude surgery. Abnormal adrenal → treat as involved.
- **Pathological stage** = most important factor; then grade, size, subtype, necrosis, sarcomatoid. Node-positive 5-y 5–30%. Thrombus with otherwise confined disease: 5-y 45–69%. Venous wall invasion as important as level; sinus fat possibly riskier than perinephric; collecting-system invasion adverse.
- Adverse clinical: symptoms, weight loss >10%, poor PS, anaemia, hypercalcaemia, ↑ ESR/LDH/ALP, thrombocytosis.
- ASSURE post hoc: algorithms barely beat stage + grade → AUA risk groups. † **UISS** = TNM + grade + ECOG PS; **SSIGN** (Mayo) = Stage, Size, Grade, Necrosis; **Leibovich** = pT, pN, size, grade, necrosis → recurrence in clear cell.

7 · Management of localised RCC

- P1 P2 Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy; Nephron-sparing surgery / partial nephrectomy in RCC — *imperative/relative/elective PN, AUA RN rule (GFR >45), surgical vs medical CKD, PN steps, enucleation, solitary kidney, AS/TA criteria, 3-cm rule, EORTC 30904, DISSRM*

AUA 2021 principles (beyond topics)

- Nephrology referral if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30. LND for clinically positive nodes; adrenalectomy only if suspicious; examine adjacent parenchyma for medical renal disease.
- ~25% of "ideal" PN candidates already have CKD; CKD after RN 65% vs PN 20%. Surgical CKD ~0.7%/y vs medical ~4.7%/y. PN survival advantage in retrospective series = selection bias. Hypertrophy 20–30%. **Predicted GFR after RN ≈ global GFR × contralateral split function × 1.25**.
- AS: small masses grow 0.1–0.3 cm/y, metastasis <2%; imaging every 3–6 mo (AS) or 6–12 mo (expectant, often palliative). **Triggers**: >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms; a fast-growing subgroup may lose the PN/TA window.
- PN: function = parenchymal quality (fixed) → **vascularised parenchyma preserved (primary, modifiable)** → ischaemia (secondary). Leak and bleeding commoner than RN; lap/robotic ≈ open. Recurrence 1–2% balanced by 1–2% contralateral RCC. Steps → *Ch 142*.
- Solitary kidney: ESRD ~4.5%; need ≥20–30% of one kidney; ACE inhibitor + low-protein diet after >50% loss. Bilateral, PN impossible: RN + dialysis (transplant after cancer-free interval) or neoadjuvant TKI (stop several half-lives around surgery).
- Enucleation controversial in sporadic RCC: 40–50% infiltrate pseudocapsule (rarely low grade, chromophobe, oncocytoma) → RMB helps; report by SIB protocol.

Radical nephrectomy

- Robson: early artery then vein ligation, outside Gerota's, adrenalectomy, extended LND. Today: perifascial dissection essential (~25% of cT1b/T2 invade fat); **no routine adrenalectomy** (unless enlarged on imaging or extensive tumour) and **no routine extended LND**.
- Open: subcostal (preferred), thoracoabdominal (huge upper pole). Laparoscopic RN standard without gross invasion, major venous involvement or bulky nodes — but has driven RN overuse → *Ch 141*.

Thermal ablation → *Ch 143*

- Best <2.5–3 cm; hydrodissection near bowel/hilum; no pathological staging.
- Cryoablation: lethal ≤−20 °C, double freeze–thaw, iceball ~1 cm beyond tumour; neither modality superior. Recurrence (table below) mostly salvaged by repeat TA; no difference in metastasis or CSS. Nodular enhancement = recurrence.

Familial RCC — 3-cm rule (NCI)

- VHL tumours <3 cm: no metastases at ~5 y; ~25% metastasise when larger; **HLRCC is the exception**. Recurrence near-universal (cyst linings) → lifelong surveillance. **Exclude phaeochromocytoma** and CNS haemangioblastoma first. Transplant after removing all renal tissue safe.

8 · Locally advanced RCC

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Clamp infrarenal IVC → contralateral renal vein → suprarenal IVC; occlude lumbars
III	Intrahepatic, below diaphragm	T3b	Mobilise liver, milk thrombus below hepatic veins; or suprahepatic clamp + Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass ± hypothermic arrest; avoid opening heart if thrombus can be brought below atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins, graft/reconstruct

- Image close to surgery (thrombus grows). **No routine renal artery embolisation** (more deaths). Coronary angiography before bypass (III–IV); intra-op TOE.
- Occluded IVC may need no reconstruction (collaterals); bland distal thrombus → remove, ligate or filter.
- Level and wall invasion predict survival; level IV without metastasis curable. Supradiaphragmatic mortality 5–10%.

Locally invasive (T4)

- Only ~40% of suspected invasion confirmed. En bloc resection is the only cure; prognosis poor. Debulking rarely (1-y survival 10–20%).

Lymph node dissection

- Only 30–43% of enlarged nodes malignant. Risk factors: ≥10 cm, cT3/T4, grade 3–4, sarcomatoid, necrosis; ≥2 → >40% node-positive.
- No survival benefit (EORTC 30881) → LND for suspicious nodes, mainly staging; resect bulky nodes if feasible.

9 · Recurrence, adjuvant therapy & follow-up

Local recurrence	After RN	After PN	After TA
Rate	2–4%	1.4–10%	Cryo 3–10%, RFA 5–20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone
Management	Only 20–40% isolated → full work-up; en bloc resection → 30–50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- Positive PN margin: usually surveil; completion nephrectomy only for gross involvement.
- Adjuvant rationale: 20–35% metastasise after "curative" surgery. Failed: cytokines, vaccines, girentuximab, most TKIs. **Adjuvant pembrolizumab** FDA-approved for high risk. Neoadjuvant TKI: 10–30% shrinkage; ~20% serious events.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1–2	X-ray
Intermediate	pT1 grade 3–4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging from 3–6 mo, rising with risk, very high up to 10 y; microscopic positive margin → up one level. **30% of recurrences after 5 y**.
- Benign pathology → no imaging; after TA: contrast imaging within 6 mo, then intermediate schedule.

10 · Other renal malignancies

Tumour	Clues	Management
Sarcoma (1–2%)	**Leiomyosarcoma commonest** (50–60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins — first resection best chance; margin + grade = prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%); massive adenopathy, splenomegaly; prior lymphoma → recurrence until proven otherwise	**Biopsy, avoid nephrectomy**; R-CHOP ± radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung commonest; multifocal, 5–30 HU	Biopsy; systemic; nephrectomy only for refractory bleeding
Carcinoid	**Horseshoe kidney (82×)**	Excision; good prognosis
Small cell	Usually advanced	Surgery + platinum
PNET	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than children

- Sarcoma: any RCC feature excludes it (vs sarcomatoid RCC); haemangiopericytoma vascular → embolise pre-op. Leiomyosarcoma > liposarcoma (reverse of retroperitoneum). Pre-op radiotherapy may reduce positive margins.
- Lymphoma: ↑ with immunosuppression; renal failure more from hypercalcaemia/urate than replacement.

> **KEY POINT**
> Non-RCC malignancies are almost all **hypovascular**; key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis — biopsy, don't resect). RCC nodes: interaortocaval (right), para-aortic (left); nodes elsewhere suggest lymphoma. A solitary enhancing mass years after organ-confined non-renal cancer is more likely primary RCC.

11 · Landmark trials

Trial	Design	Result	Take-home
EORTC 30904	RCT RN vs elective PN, ≤5 cm, normal contralateral	PN better function; 10-y OS better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed

Trial	Design	Result	Take-home
EORTC 30881	RCT RN ± LND	No survival benefit	Underpowered, low-risk (4% node+); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	DFS 6.8 vs 5.6 y; no OS gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	DFS HR 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

1 · Epidemiology & risk factors

- P3 P2 Upper Tract Urothelial Carcinoma: Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management; Etiopathogenesis and management of Upper tract TCC — *epidemiology, tobacco/aristolochic acid/Lynch/occupation/phenacetin/arsenic, seeding vs field effect, CTU, cytology, URS biopsy, TNM, EAU risk groups, management*
- UTUC = **5–10%** of urothelial cancers; Balkans **40%** of renal neoplasms. Bilateral ≤5%; multifocal **10–20%**; concomitant CIS 11–36%; concurrent bladder cancer **17%**. ~60% invasive, 7–9% metastatic at diagnosis.
- After UTUC: bladder recurrence **22–47%**, contralateral upper tract 2–6%.

KEY POINT

Why UTUC presents invasive: thin (mesodermal) muscle layer → reaches muscle, parenchyma or periureteric fat earlier than bladder tumours.

- **Smoking**: RR **2.5–7**, ureter > pelvis; stopping >10 yr cuts risk 60–70%. Occupational exposure ≈7 yr. Chronic infection/stones RR 1.5–2 (SCC). Iatrogenic: cyclophosphamide (acrolein), anthranoid laxatives >1 yr, pelvic radiotherapy.
- Aristolochic acid: Belgium 1992–93 slimming herbs; no sex bias, ≈10 yr earlier; TP53 codon 139. Arsenic: female 2:1, ureter 2:1; causal link unproven. Phenacetin: latency ≈22 yr; pelvis > ureter. Lynch: mean age 55, relatively more women, upper tract > bladder.

KEY POINT

All four behave as a **field disease** (multiple, bilateral recurrences) → kidney-sparing more attractive, but BEN patients often have renal insufficiency. Analgesic abuse history in 22% of pelvic and 11% of ureteric tumours.

- **Lynch**: UTUC risk ≈**14×**, ≈**75×** in MSH2. **EAU: screen at first interview** if age <60, personal LS-spectrum cancer, FDR <50 with LS cancer, or two FDRs with LS cancer → germline sequencing → surveillance for colorectal/endometrial cancer, family counselling. MMR loss on IHC → germline testing.

2 · Pathology & molecular biology

- >90% UC; variant histology ≈**25%** (squamous, glandular, sarcomatoid, micropapillary, neuroendocrine, lymphoepithelial) = aggressive; **micropapillary** independently adverse. CIS easily missed.
- Ureteric inverted papilloma: malignancy ≈**18%** → follow ≥2 yr.
- SCC 0.7–7%, 6× commoner in pelvis, invasive at presentation. Adenocarcinoma <1%, stones/obstruction, poor prognosis. Sessile tumours invade in >80%.
- UTUC and bladder cancer genetically distinct but clonally related. **RAS-mutated** subtype worst. MSI and hypermethylation distinguish from bladder; MSI prognostic and flags LS. Uroplakin III and p27 expression favourable.

3 · Diagnosis

- Haematuria commonest; flank pain; mass ≈10%; incidental ≈15%. **Cystoscopy always**; positive cytology with normal cystoscopy (incl. urethra) → evaluate upper tracts.
- **CTU** first line (EAU prefers to MRU): sensitivity ≈**92%**, specificity ≈95%; filling defect 50–75%; misses flat lesions; UC ≈46 HU vs uric acid stone ≈100 HU. Hydronephrosis = invasive ureteric tumour in up to 80%.
- MRU: avoid gadolinium if CrCl <30 mL/min. Retrograde pyelography ≈75%. Split-function scan before radical surgery.
- Cytology ≈65%: G1 20%, G2 45%, G3 75%; selective washing > voided; brush + cytology ≈90%; **collect before contrast**. FISH best for low-grade surveillance, not routine.
- Flexible URS + biopsy: accuracy 75–85% → ≈90%; biopsy grade ≈90%, visual grading only ≈70%.

> **KEY POINT**
> **Why URS biopsy cannot stage:** small, shallow samples → Ta biopsy often T3 at resection. Stage is **inferred** from grade, appearance, hydronephrosis and cytology → kidney-sparing patients need intensive follow-up.

- **Diagnostic URS indications:** uncertainty; kidney-sparing planned; solitary kidney. **Pitfall:** URS before RNU ↑ intravesical recurrence (esp. pelvic tumours, biopsy), survival unchanged → avoid when CTU + cytology already justify RNU.

4 · Staging & grading

TNM 8th	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or **renal parenchyma**; ureter: periureteric fat. pT3a = microscopic parenchymal; pT3b = macroscopic or peripelvic fat (worse)
T4	Adjacent organs, or through kidney into perinephric fat
N1 / N2	Single node ≤2 cm / single >2 cm or multiple

- AJCC: I = T1; II = T2; III = T3; **IV = T4, any N+ or M1**. Nodes: pelvis/upper ureter → hilar, retroperitoneal; lower ureter → pelvic. WHO 2004/2016: PUNLMP (managed as low grade), low, high grade.

5 · Prognosis & risk stratification

- **Stage most important**, then grade. SEER 5-yr CSS after RNU: ≤T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- **Ureteric and multifocal tumours do worse** than pelvic at the same stage (no parenchymal barrier; periureteric lymphovascular plexus).
- Pre-op: age (never alone a reason to deny RNU), smoking, ASA/ECOG, obesity, low albumin, high NLR/CRP; **RNU within 12 wk** for invasive disease. Size **2 cm** best cut-off for ≥pT2.

- Post-op: nodes (up to 40%), extranodal extension, node density ≥30%, LVI (≈20%; never on biopsy → report on every specimen), margins, necrosis >10%, sessile, CIS, variant histology.

Low risk — ALL	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
<2 cm	>2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

KEY POINT

Every pre-op factor is a stand-in for invasion; one adverse feature = high risk. **Trap:** small, low-grade, unifocal tumour **with hydronephrosis is high risk**.

6 · Recurrence patterns

- Bladder after UTUC: 15–75% at 5 yr, usually within 2 yr.
- Upper tract after bladder cancer: 2–4%, late (median >3 yr); 0.75–6.4% after cystectomy; ≈**30%** of BCG-treated bladder CIS (or cystectomised for BCG-refractory disease) develop UTUC. Risk: distal ureter/orifice tumour, multifocal/recurrent NMIBC, CIS, reflux.

KEY POINT

Asymmetric recurrence: antegrade seeding → bladder recurrences common, early, monoclonal; field effect → multifocal tumours of different clones. Both operate.

- Intravesical recurrence after RNU — patient: male, prior bladder cancer, smoking, CKD; tumour: positive cytology, ureteric, multifocal, invasive, necrosis; treatment: laparoscopic, extravesical cuff, positive margins, diagnostic URS.
- **Pan-urothelial disease** = bladder + ≥2 extravesical sites (women: both upper tracts; men: upper tract(s) and/or prostatic urethra). No evidence-based algorithm: cystectomy for high-grade bladder disease, conservative for non-infiltrating UTUC, total urinary tract removal in the young. Cystectomy then RNU may lose cisplatin eligibility.

7 · Management overview & follow-up

ALGORITHM

Cystoscopy + CTU ± URS biopsy/cytology + contralateral function → LS screening → risk-stratify

Low risk → kidney-sparing (endoscopic or segmental) + intensive surveillance

High risk → RNU + bladder cuff ± LND, within 12 wk

cN+ → systemic chemotherapy first → surgery if response

After RNU → single intravesical chemotherapy dose → cystoscopic follow-up for all

- Pre-op nomograms (≥pT2/non-organ-confined) select for neoadjuvant chemotherapy; pathological downstaging improves OS. Chemotherapy before RNU because nephrectomy often removes cisplatin eligibility. LND gives best staging; curative role debated. Technique and POUT → *Ch 139*.

- Follow-up: cystoscopy for all; after kidney-sparing, URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory): yearly upper-tract CT (replaces cytology and excretory urography); risk constant → prolonged surveillance.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Epidemiology, risk factors, pathology, staging and imaging → *Ch 138*

1 · Principles & patient selection

- P3 Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management — *AUA/SUO favourable/unfavourable groups and ablation role, elective vs imperative KSM, URS and percutaneous steps, lasers, segmental options, RNU principles, abandoned cuff techniques, topical BCG/UGN-101, follow-up*
- **Standard**: RNU with bladder cuff if contralateral kidney normal and patient fit. UTUC is **unforgiving of spillage** → every approach keeps the collecting system closed.
- Work-up: CT/MR urogram → cystoscopy → URS + biopsy + selective cytology (accuracy ≈75% → 85–90%). No retrograde access (diversion) → antegrade endoscopy; endoscopy unsafe → image-guided needle biopsy. Keep diagnostic endoscopy brief and low-pressure (seeding, pyelovenous/pyelolymphatic migration).
- LG tumours mostly Ta/T1, HG mostly invasive.

AUA/SUO risk group	Features	Ablation	Systemic therapy
Low – favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	**Preferred**	No
Low – unfavourable	LG + obstruction, multifocality or prior lower-tract tumour	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High – unfavourable	HG cytology, radiographic invasion, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

KEY POINT
Low grade ≠ automatic ablation: obstruction or multifocality makes an LG tumour unfavourable, where RNU keeps a role.

KEY POINT
Two axes — tumour risk and renal reserve. High risk + adequate reserve → extirpation; low risk → preservation; imperative indications are where they conflict.

ALGORITHM
Imaging suspicion → CT/MR urogram → endoscopy + biopsy + cytology → risk group + shared decision

High risk + adequate GFR → RNU ± neoadjuvant chemotherapy · Low risk or poor GFR → kidney-sparing

- Imperative KSM must not significantly compromise cancer control; weigh dialysis vs oncological risk. Lower ureteric (esp. unfavourable) → distal ureterectomy, not endoscopy.
- **Expectant management** (AUA/SUO option): only with major comorbidity, competing mortality or high ESRD risk; untreated UTUC progresses with markedly worse survival.

2 · Endoscopic (kidney-sparing) management

- **Grade** is the key selector and strongest recurrence predictor; HG results poor. AUA/SUO: tumours **<1.5 cm**; caution ≥1.5 cm.
- **Staged approach** for bulky/suspected HG: biopsy + partial ablation only (avoid perforation before possible RNU); complete once pathology confirms.
- Vs RNU: similar survival but **~3× more reinterventions**; lifelong surveillance. Papillary/fornix tumours hard to treat safely.

	Retrograde ureteroscopic	Antegrade percutaneous
System	**Closed**	**Open** – tract seeding ≈0.3%, all HG
Best for	Low-volume ureteric, small renal	Large pelvic/upper ureteric, lower-pole calyx, diversion
Staging	Not possible	Possible – base sent separately
Morbidity	Low	Higher (≈25%), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical Rx	Repeat URS; catheter or stent reflux	Via retained nephrostomy

- **URS**: saline barbotage improves cytology; tight ureter → pre-stent (passive dilation), not forceful dilation; access sheath — no recommendation. Fix minute specimens at once; selective cytology + biopsy improves grading.
- Techniques: forceps/basket excision + base ablation; ureteroscopic resection never beyond lamina propria; usual: **3-Fr biopsy then laser**. Orifice-protruding tumour → ablate or resect whole intramural ureter. Bugbee depth unpredictable; **never fulgurate circumferentially**.
- Lasers: Ho:YAG <0.5 mm (ideal for ureter); Nd:YAG 5–6 mm (transmural risk); thulium <0.2 mm (strong coagulation).
- **Percutaneous**: puncture calyx distal to a peripheral tumour, never through it; upper/middle calyx for pelvis. 30 Fr (tract may mature 1–2 weeks). Cold-cup (thin stalk) or **bipolar** resectoscope (monopolar → hypo-osmotic absorption). **Nephrostomy always left** → second look, topical therapy. HG/invasive → RNU.
- Diversion: know anastomosis type; flexible endoscopy first, else antegrade wire into conduit then retrograde URS over it.

> **KEY POINT**
> Does URS spread tumour? Prior ureteroscopy ↑ bladder recurrence (seeding), but pre-RNU URS has not ↑ metastasis. Mitigate with single post-op intravesical instillation.

3 · Segmental ureterectomy

- Saves nephrons **and** gives a staging specimen; in selected patients equals RNU (stage/grade set prognosis). **URS of the entire tract above is mandatory**. HG → ipsilateral pelvic LND.
- Ipsilateral recurrence **33–55%**, mostly distal to the original site, lifelong.
- Segmental resection + uretero-ureterostomy (rare): LG proximal/mid tumour not ablatable, or HG when unit must be saved; 1–2 cm margins, frozen section, spatulate, stent, tension-free.
- Distal ureterectomy + reimplant: LG low-stage, or selected HG distal when renal preservation needed; whole distal ureter + cuff ± LND. Non-refluxing (limits seeding) vs refluxing (eases surveillance) debated.

- **Boari flap**: small/irradiated bladder contraindicates; base ≥2 cm wider than apex; width ≥3× ureteric diameter. **Ileal ureter**: 20–25 cm ileum ≥15 cm from ileocaecal valve, isoperistaltic; Foley ≥1 week; cystogram + nephrostogram before removal.

Manoeuvre	Length gained	Condition
Psoas hitch	≈5 cm	Mobile bladder; avoid nerve entrapment
Boari flap + hitch	Further 10–15 cm	Adequate capacity
Renal descensus + nephropexy	Up to 8–10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 · Positive cytology & carcinoma in situ (CIS)

ALGORITHM

Positive voided cytology → repeat → CT urogram + bladder biopsies → bladder positive: treat bladder

Persistent positive, negative bladder (or after BCG "6+1") → selective upper-tract cytology + RPG + prostatic urethral biopsy (men) + URS

- Upper-tract CIS = exclusion diagnosis (persistent positive selective cytology, normal URS/imaging). **No RNU for cytology alone** (false positives from infection/stones; contralateral disease follows); segmental resection fails (field change).
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy; late recurrences can be lethal. RNU if more than surface disease. CIS with HG papillary UTUC → radical/multimodal.
- CIS at ureteric margin during cystectomy: frozen section; conversion to negative margin achieved in most, better survival.

5 · Topical therapy in the upper tract

- Indications: solitary kidney, bilateral UTUC, ESRD risk. Routes: nephrostomy (≤25 cm H_2O), ureteric catheter, stent reflux. **Always low pressure, sterile urine** (exclude infection before every instillation) — sepsis deaths reported.
- Agents as in bladder; evidence small uncontrolled series. Adjuvant instillation after papillary ablation: **no proven benefit**; BCG acceptable after complete ablation of HG favourable Ta/T1. BCG complication: renal granulomas (commonest).
- **UGN-101 (Jelmyto)**: MMC reverse-thermal gel; FDA 2020 for LG UTUC **not endoscopically manageable**; stricture ≈44% (fewer antegrade), marrow suppression, embryo-fetal toxicity. Complete endoscopic ablation remains mainstay.

6 · Radical nephroureterectomy

- **Indications**: non-metastatic high-risk pelvic/proximal ureteric tumour (HG, obstruction, parenchymal invasion) with cuff + LND; LG with unfavourable features or failed/infeasible endoscopy.
- **Kidney first, cuff last**; **clip ureter early**; hilum before manipulation; outside Gerota's. **Adrenalectomy not routine**. Never seal a divided ureter with a vessel-sealer alone. Midline best for nodes and bladder; flank incisions need a second incision for the cuff.
- Remove **entire intramural ureter and orifice**: retained stump recurs in **30–75%**, can't be surveyed, worsens survival. After bowel diversion, excise ureteroenteric anastomosis with

bowel cuff.

- Cuff: extravesical or transvesical (1-cm mucosal cuff) acceptable; lasso, pluck, stripping, laparoscopic staple abandoned (spillage, recurrence, positive margins). Two-layer closure; Foley several days.
- Laparoscopic/robotic: Foley before insufflation; extract **intact in a bag**; wide perihilar margin. Ureteric blood supply: anteromedial (upper ⅓), medial (middle ⅓), **lateral** (lower ⅓). Relative contraindications: bulky invading tumour, post-chemotherapy LND.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

KEY POINT
Principle beats approach: port-site seeding and carcinomatosis follow spillage, not the approach.

Lymphadenectomy

- P4 Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — *UTUC templates (right interaortocaval), ≥8 nodes, chylous leak; neoadjuvant rationale, EA8141, AUA/SUO CrCl <50, POUT*
- Value in **T2–T4**, esp. pelvis/proximal ureter; surgeon, not approach, sets yield; pN0 > pNx > pN+. Caudal limits: renal pelvis → IMA; upper ⅔ ureter → aortic bifurcation; lower ⅓ → pelvis.
- Prognostic cliff **T2 → T3**; HG behaves like T3; N+ with surgery alone ≈ metastatic disease. Renal function loss is RNU's main cost.

7 · Perioperative adjuncts

- Bladder tumours in 30–50% after RNU/endoscopy, mostly within 2 yr. **AUA/SUO strongly recommend** single dose after RNU (optional after endoscopy). Post-op (3–7 d): cystogram → MMC or gemcitabine, dwell 1–2 h. **Intraoperative** option: instil before surgery, clamp 1–2 h, drain once ureter clipped — no cystogram, drug present when seeding occurs.
- **Renal function is the hinge** → platinum **before** nephrectomy. Neoadjuvant favoured if post-op CrCl expected **<50** mL/min; EA8141: CrCl >50 → accelerated MVAC ×4. Adjuvant (POUT): gemcitabine–cisplatin (CrCl >50) or –carboplatin (30–49); cisplatin preferred. cN+ → chemotherapy, then consolidative surgery.

8 · Complications

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; no circumferential fulguration
Percutaneous	Bleeding, irrigant absorption, perforation, secondary UPJ obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure

Procedure	Complication	Management / prevention
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 · Follow-up

- Every visit: **cystoscopy**, cytology, metabolic panel; after KSM add ipsilateral URS; don't routinely scope normal contralateral tract. CT/MR urography (RPG if no contrast); filling defects → URS; chest imaging; smoking cessation.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6–9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3–6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	+ cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3–6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3–6-monthly ×2 yr; chest CT 6–12-monthly ×5 yr

10 · Landmark trials

Trial	Design	Result	Take-home
ODMIT-C	RCT, MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% → 16% at 1 yr	Single-dose instillation standard
Pirarubicin trial (Ito)	RCT, single pirarubicin dose within 5 days of RNU	Bladder recurrence 42% → 17% at 2 yr	Confirms single post-RNU instillation
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option
POUT	Phase III adjuvant platinum vs surveillance, pT2–4/N+	Stopped early; DFS HR **0.45**; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

141 Contemporary Open Surgery of the Kidney

1 · Role & preoperative work-up

- Open surgery = **exposure + early great-vessel control**: renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours, hostile abdomen → *Ch 142*.
- **Renal function**: eGFR by **CKD-EPI 2021 race-free** equation (better than MDRD) decides partial (PN) vs radical nephrectomy (RN); cystatin C detects mild GFR loss.
- GFR **<60** or significant proteinuria → perioperative nephrology consult; optimise diabetes/hypertension even if GFR ≥60.
- Advanced RCC: blood count, metabolic panel (MSKCC/IMDC risk models), liver tests + coagulation (**Stauffer syndrome** — paraneoplastic hepatic dysfunction), cardiopulmonary fitness.
- Contrast generally safe if GFR **>30**. Suspected thrombus → MR or CT venography **≤1 month old** (rapid propagation).
- Prophylaxis: cefazolin (or clindamycin) ≤1 h pre-incision; no routine bowel prep; pneumatic compression + heparin 5000 IU SC; stop smoking 4–6 wk.
- **Renal artery embolisation**: allows vein-before-artery ligation when hilum hidden; no longer routine: postinfarction syndrome (pain, fever ≤72 h); may *increase* blood loss; harder if surgery delayed >48 h. If used, embolise evening before/morning of surgery and confirm inflow occluded.
- Atraumatic vascular clamps (bulldog, Satinsky) — never Kocher on vessels; argon beam for liver tears.

2 · Surgical anatomy & approaches

- At risk: **right** — liver, duodenum, colon; **left** — spleen, pancreatic tail, colon, superior mesenteric artery (SMA).
- **Right renal artery**: posterior-inferior to renal vein, behind IVC; reachable in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- **SMA**: anterior aorta **1–2 cm above LRV**, veers left → mistaken for left renal artery.
- Right gonadal/adrenal veins → IVC; left → LRV.
- Anterior exposure = colon off Gerota's via **white line of Toldt**; right adds **Kocher** (no cautery — duodenal injury devastating); left divides splenorenal ligament.
- **Cattell-Braasch**: Toldt → caecum → Kocher → mesenteric root to Treitz; midgut pivots on SMA, exposing aorta/IVC. **Infracolic** (trauma, bulky hilum): below Treitz → aorta + LRV. Partial **Mattox** (left): spleen and pancreas rotated medially together.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour/IVC thrombus (early aortic access); bilateral	Bowel mobilisation
Anterior subcostal	Unilateral transperitoneal	Muscle division
Chevron	Renovascular; RN + thrombectomy; both pedicles	Extensive muscle division

Incision	Best for	Limitations
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN/PN; 8-cm mini-flank supra-11th	Pleura
Transcostal (11th, rib resected)	Retroperitoneum at old rib tip	Rib division
Thoracoabdominal, dorsal lumbotomy	Largely historical; thoracoabdominal still an option for level III–IV thrombus	

- **Supine**: excess hyperlordosis → back pain (avoid in spinal stenosis). **Flank**: 12th rib tip over kidney bar, axillary roll; ↓ venous return and lung function — poor for cardiopulmonary disease, severe kyphosis; pad against rhabdomyolysis.
- **Supracostal** (popularised by Turner-Warwick): incise on upper rib border → open lumbodorsal fascia at rib tip → mobilise pleura up → divide intercostals close to **upper border of rib** (spares bundle). Spare **subcostal nerve** (between internal oblique and transversus).

3 · Simple nephrectomy

- **Within Gerota's fascia** for benign nonfunction (obstruction, infection, stones, reflux, polycystic, dysplasia); functioning kidney rarely (pain, bleeding, renovascular hypertension). Stay on the capsule; spare adrenal.
- Benign PN: focal atrophy, duplex pole, infected diverticulum, symptomatic angiomyolipoma.

4 · Radical nephrectomy

- Robson (1969): kidney + Gerota's + adrenal + nodes crus → bifurcation. Now: **spare normal adrenal**, selective lymph node dissection (LND).
- **Indications**: not PN-amenable — nonfunctioning kidney, tumour replacing kidney, nodes, cytoreduction, venous thrombus.
- Staging: abdominal CT/MRI + **chest CT**; bone scan only for symptoms, fracture, suspicious imaging or raised bone alkaline phosphatase; no routine brain imaging.
- Biopsy only if it changes plan: infiltrative mass (urothelial, lymphoma, metastasis), before systemic therapy, bilateral, trials.
- T4 invasion = altered organ enhancement, not contact. Liver rarely truly invaded (depth usually <1 cm).

> **ALGORITHM**
>
> **Right:** survey abdomen → Toldt, iliac → hepatic flexure → Kocher → IVC → loop and palpate renal vein → artery behind vein → **artery then vein** → kidney outside Gerota's.
>
> **Difficult hilum:** interaortocaval artery ligation → kidney decompresses → flaccid vein checked, double-ligated (0 silk + 3-0 Prolene) → lateral artery divided. Or: lower pole → ureter and gonadal vein → divide ureter → lift kidney posterior to anterior to show artery.
>
> **Left:** Toldt splenic flexure → iliac → renocolic ligament (pad pancreas) → aorta leads to LRV; gonadal vein leads up to it → artery above/behind LRV → 0 silk + 3-0 Prolene, confirm occlusion → vein.

KEY POINT

Why artery first? Vein first leaves kidney engorged and collaterals bleed. Arterial ligation shrinks the kidney and softens the vein so it can be checked for thrombus. Persisting pulse = accessory artery, incomplete control, or SMA tied.

- **En bloc pedicle** ligation (hilar nodes): arteriovenous fistula (AVF) theoretical, very rare.
- **Lumbar veins**: silk ties; **no clips on IVC side** (block emergency caval clamping).
- **Loss of pedicle control**: tell anaesthetist, finger/sponge compression, 2 suckers, **never clamp blindly**.
- **Adrenalectomy** only for upper-pole tumour with lost plane, adrenal mass, diffuse involvement. Metastasis **<5%**; CT/MRI miss 20–25%. Contralateral metastasis as common as ipsilateral → prophylactic removal risks adrenal insufficiency without benefit.
- **LND**: nodes positive ≈5%. EORTC RCT (772 patients): no overall survival benefit, but many low-stage tumours. **Do if nodes enlarged** (short axis **≥1.5 cm ≈ 50% positive**); consider in high risk (cytoreduction, >10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- Templates: right — paracaval, precaval, retrocaval, interaortocaval; left — para/retroaortic; crus → bifurcation. **Clip lymphatics above renal vein** → prevents chylous ascites (cisterna chyli).
- **Closure**: saline + big breath; bubbles = pleural tear → suture over 14-Fr catheter, withdraw on held breath; **chest X-ray in all**.

5 · IVC tumour thrombectomy

- RCC = commonest cause; **5–10%** of RCC. Untreated → most die within 1 yr; surgery (RN + LND ± systemic therapy) is the backbone. Others: Wilms, adrenocortical carcinoma.
- **Tumour thrombus** vs **bland thrombus** (clot below obstruction) — managed differently. **MR venography** best for extent, wall invasion, bland thrombus.
- Preop: screen for PE/DVT; anticoagulate if VTE, bland thrombus or IVC occlusion. **No suprarenal filters** (thrombus grows onto device). Embolise only for unexcisable hepatic-vein thrombus or bleeding kidney.
- **Transoesophageal echo (TEE)** for level II–IV. Vascular/liver surgeon II–III; **cardiothoracic for III–IV**.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC **≤2 cm** above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At/above major hepatic veins, below diaphragm	Liver mobilisation ± Pringle; occlusive → suprahepatic clamp; nonocclusive → venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) + deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip/staple); don't extract

Bland group	Location	Action
D	Mixed with tumour	Resect infrarenal IVC; keep lumbars as collaterals

ALGORITHM

Level I (right): control IVC before handling tumour → right artery ligated interaortocavally → loop LRV, suprarenal and infrarenal IVC → **Rummel tourniquets** → milk thrombus toward ostium → Satinsky around ostium (thrombus inside jaws) → ostial cavotomy keeping ≥half IVC width → extract.

Level I–II: early renal artery ligation → tourniquets **infrarenal IVC → contralateral renal vein → suprarenal IVC** → cavotomy → extract → heparinised saline flush, inspect wall (resect suspicious intima) → 4-0 Prolene → release **caudal clamp first**, flush 5–10 mL on held breath → cranial clamp.

- **Left tumour (level II)**: harder — IVC reached from right, both colons mobilised. Left artery first, then Kocher, **split-and-roll** IVC, excise LRV ostium en bloc. Don't clamp right renal artery (no venous collaterals); intermittently release right renal vein. No routine heparin.
- **Level III**: bypass on standby; transplant-style liver mobilisation. **Major hepatic veins never divided**; accessory (caudate lobe) veins tied → 2–3 cm more IVC. Level III is defined by major, not accessory, hepatic veins. **Budd–Chiari = contraindication**.
- **Pringle** (clamp portal vein, hepatic artery, bile duct) if clamping above hepatic veins: ≤60 min, ideally **≤20 min**. Pringle + clamps above and below liver = total hepatic vascular occlusion.
- **Level III sequence**: clamp infrarenal IVC → contralateral renal vein → porta hepatis → suprahepatic IVC → cavotomy → extract (Fogarty or 20-Fr Foley for unreachable thrombus); unscrapable wall invasion → IVC resection.
- **Clamp trial** 2–5 min: preload ↓60%, cardiac output (CO) ↓40%. **CO ↓>50% or MAP ↓>30%** → bypass or supracoeliac aortic clamp. Occlusive thrombus tolerates clamping (azygos collaterals); free-floating does not. Also bypass for cardiac/hepatic dysfunction, poor contralateral kidney, portal hypertension, uncontrolled bleeding.
- Incomplete tumour removal is universally fatal → never compromise IVC control to save time.

	VVB	CPB + DHCA
Access	Femoral → jugular, percutaneous	Sternotomy; 16–18 °C, drain 95% blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar/intercostal back-bleeding	Cardiac, neurological
Use	Least invasive; not with atrial or iliac bland thrombus, Budd–Chiari	Level IV; bloodless field; ≤60 min (<40 better)

- **VVB**: cannula tips far from thrombus (dislodgement → massive PE). **DHCA**: when thrombus cannot be milked below an intrapericardial clamp.
- P3 Vascular grafts and vascular surgery in urology — *caval repair: ≥50% lumen primary closure, else pericardial patch (Leibovich–Bower width), circumferential → PTFE 16–20 mm + lifelong aspirin 81 mg; left tumour needs right-kidney drainage*
- Patch: 5-0 Prolene, small regular bites. Suprarenal IVC ligation with right tumour OK if LRV tied caval to its tributaries (collaterals).

KEY POINT

"Shortcuts" that become "longcuts": opening IVC before all inflow controlled; side-biting a level I; clips on IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping test clamp.

6 · Partial nephrectomy

- P2 Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches — *PN standard for T1; imperative/relative/elective; relative contraindications (<20% nephron mass, pedicle encasement, thrombus, cT4, nodes); hereditary observe to 3 cm except HLRCC*
- PN size-independent if safe; collecting-system invasion also a relative contraindication. **Hyperfiltration** (FSGS, proteinuria, CKD) rare unless nephron mass ↓>80%.
- Multifocality: clear cell ≈2%, papillary ≈10% → inspect whole kidney; intraoperative ultrasound (IOUS). Hereditary: preserve and reapproximate perirenal fat/Gerota's to ease later surgery.

Question	Evidence
Warm ischaemia	Aim **<30 min**; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss, stay

KEY POINT

PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, **volume of parenchyma preserved** matters more than clamp minutes; hypothermia and mannitol optional (role in solitary kidney/CKD unclear).

- Always **control hilum first**, even if not clamping. Options: manual compression, parenchymal clamp, selective branch, whole pedicle, early unclamping, off-clamp. Leaving vein open: no proven benefit.
- **Enucleation**: pseudocapsule plane, minimal normal tissue. **Resection**: 5–10 mm rim, usually clamped; more small vessels transected. Frozen section of deep margin if needed.
- **Renorrhaphy**: inner — 4-0 absorbable to vessels, running collecting-system closure (no permanent sutures = stone nidus); outer — capsular sutures over bolster. Drain if collecting system entered.

7 · Complications & management

Injury	Management
SMA	Mistaken for left renal artery, Cattell-Braasch traction. Potentially lethal → vascular surgeon, primary repair or graft
IMA/IMV	Ligation safe if marginal artery of Drummond patent
Superior mesenteric vein (SMV)	Ligation → bowel ischaemia, splanchnic hypertension, hypotension. **Never ligate**; 6-0 Prolene venorrhaphy; transection → reanastomosis or autologous vein; leave abdomen open
Duodenum	Haematoma narrowing lumen → incise serosa/muscularis, drain; tear → debride, 2-layer closure, omentum, drain, NG tube

Injury	Management
Pancreas	Glue + drain (amylase); ductal → general surgeon; late collection → percutaneous drain, nil by mouth, octreotide
Pleura	Purse-string over 14-Fr catheter, evacuate on positive-pressure breath; large → chest tube

- **Thrombectomy**: air embolism (release caudal clamp first, let ≈5 mL blood escape); PE (pulmonary arteriotomy if collapse); bleeding despite control = untied lumbar; **no cell saver** (spreads tumour); liver enzymes peak 2–3 days.
- **Urine leak** (large, endophytic, hilar tumours; most within 72 h): drain out by 48 h if low. Output >50 mL/24 h → drain creatinine > serum = urine. Persisting urinoma >48 h or symptoms >24 h → retrograde double-J stent + Foley. Most resolve 4–6 wk.
- **Pseudoaneurysm** (flank pain, haematuria, shock): stable → bed rest, serial haemoglobin; significant → **angiography** (CT angiography misses it, can't treat) + **selective embolisation**; refractory → re-explore.
- **AVF**: bruit, haematuria, heart failure, steal ischaemia; Doppler/CT/MR angiography → observe → coil embolisation → surgery if complex.
- **Renal insufficiency**: usually mild, transient (solitary kidney, large resection, long ischaemia); exclude obstruction, drug toxicity, thrombosis.

142 Laparoscopic and Robotic Surgery of the Kidney

1 · Principles & patient selection

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ; first laparoscopic PN: Winfield 1993. Minimally invasive surgery (MIS) vs open: less pain and blood loss, quicker convalescence, **equivalent long-term outcomes**. Indications same as open.
- **General anaesthesia mandatory.** CO2 pneumoperitoneum impairs ventilation, venous return, cardiac output; high risk in COPD, intracranial pathology, cardiac disease. Severe COPD (hypercarbia): lower pressure → helium → convert; AirSeal reduces CO2 load.
- Robot: **docking/undocking delays emergency access**; console surgeon may miss patient slipping → close team communication.
- Obesity: **visceral adipose tissue**, not BMI, predicts difficulty. Age no barrier. Prior surgery alters trocar sites and TP vs RP choice — **not a contraindication**.

> **KEY POINT**
> The patient must tolerate the **pneumoperitoneum**, not the renal operation — cardiopulmonary reserve, not age or BMI, decides fitness.

2 · Access & port placement

Transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted (HAL), laparoendoscopic single-site (LESS), NOTES.

	Transperitoneal	Retroperitoneal
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	After colon reflection	Direct, esp. posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- **TP:** modified flank 30–45°; 3–5 trocars (umbilical camera, subxiphoid-midline and AAL 12 mm stapler port); obese → shift lateral. **Robotic** arm spacing **8 cm (Si) / 6 cm (Xi)** + 12 mm assistant port. Ports form a **broad-based triangle** with camera at apex.
- **RP:** full flank, table flexed; incision posterior axillary line between 12th rib and iliac crest → finger dissection (psoas, lower pole) or balloon (600–800 mL) → blunt-tip cuffed trocar.

> **KEY POINT**
> RP pitfalls: balloon within muscle → flank hernia; too anterior → peritoneal/colon injury; too posterior → quadratus/psoas bleeding. Bail-out: open peritoneum, continue TP.

- **HAL:** tactile feedback, manual retraction; novices, donors, scarred kidneys (XGP). Drawbacks: forearm paraesthesia, wound infection/hernia; bag specimen (hand-port metastases). Emergency: hand through extended port compresses bleeding vessel.
- **LESS / single-port (SP) robot:** cosmesis, less pain, lost triangulation; outcomes unchanged; SP partial nephrectomy has longer warm ischaemia.

3 · Nephrectomy — operative steps

ALGORITHM

Reflect colon → find ureter/gonadal vessels on psoas → lift kidney, expose hilum → artery then vein → upper pole (adrenal spared or taken) → ureter and lateral attachments → bag → extract

- **Colon:** keep **lateral attachments** until late (else kidney rolls over hilum); plane between bright-yellow mesenteric and pale Gerota fat; close mesocolic defects. Right: cold dissection near duodenum, **Kocher**.
- **Ureter** lies just posterior to gonadal vessels (peristalsis); keep as handle; stay above psoas fascia (thigh numbness).
- **Hilum: artery first, then vein** with endo-GIA stapler; left renal vein stapled lateral to adrenal vein. **Never clips in staple line**. **En bloc hilar stapling** safe (no AVF), saves ~45 min.
- **Upper pole:** adrenal-sparing — open Gerota's above hilum, peel fat off upper pole; extreme fibrosis → subcapsular nephrectomy once vessels controlled.
- **Extraction:** intact bag, or morcellation in impermeable **LapSac** — keeps type and grade but loses size and stage.

4 · Simple vs radical nephrectomy

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple:** chronic pyelonephritis/XGP, obstructive/reflux nephropathy, TB, MCDK, renovascular hypertension, acquired cystic disease, symptomatic ADPKD. Hardest after pyonephrosis, nephrostomy, stones → hand assistance or subcapsular nephrectomy.
- **RN:** MIS = open oncologically (5/10 yr). RP vs TP RCTs: only consistent RP advantage is **shorter operative time**. Tumours >10 cm feasible (lost landmarks; ultrasound, extra ports). Conversion higher laparoscopic than robotic.
- **LND:** staging value only; no overall survival benefit; AUA: remove clinically positive nodes.
- **IVC thrombus** (experts): early arterial ligation may retract thrombus → milk back → staple or Satinsky cuff; preop embolisation helps robotic (not open) surgery *→ Ch 141*.
- **Cytoreductive**: laparoscopy → quicker recovery, earlier systemic therapy.

KEY POINT

Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery.

5 · Partial nephrectomy

- P2 P1 Nephron-sparing surgery / partial nephrectomy in RCC; Role of renal nephrometry in management of RCC — *PN default where feasible; RENAL score predicts complexity, WIT, conversion; enucleation; TP anterior vs RP posterior*
- cT1b/T2: PN better eGFR, equal oncology. **Caveat: tumours >5 cm upstaged to pT3a recur more after PN** — for large tumours PN argument is function.

- **Hilar** tumour (touches renal artery/vein) → consider enucleation; superficial renorrhaphy bites. Multifocal → selective clamping guided by 3D reconstruction. Solitary kidney: robotic PN short WIT, rare conversion. **Heminephrectomy** = ≥30% parenchyma.

ALGORITHM
Intraoperative ultrasound (size, depth) → open Gerota's away from lesion → score capsule → clamp hilum (Satinsky en bloc or bulldogs) → cold-scissor excision → close vessels/collecting system → renorrhaphy → unclamp → drain

- **Trifecta:** negative margin + no major complication within 30 days + **WIT ≤25 min**.
- **Enucleation:** less blood and function loss, more positive margins, same survival (familial, multifocal, solitary, severe CKD).
- **Renorrhaphy:** sliding-clip (Lapra-Ty), barbed sutures shorten WIT; **single-layer** → better function, fewer pseudoaneurysms. Sealants: no randomised benefit.
- **Collecting system** entered in **~60%**: 2-0/3-0 polyglactin; **always drain**; routine ureteric catheter doesn't prevent fistula.
- **Ischaemia:** historical warm limit **30 min**; **preserved parenchymal volume predicts function better than WIT**. Cold ischaemia for solitary kidney, CKD, long clamp.

Hilar strategy	Best for	Verdict
Artery-only vs artery + vein	Any	Equivalent
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (NIRF, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

KEY POINT
No clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped.

- Robotic vs laparoscopic PN: fewer conversions, shorter WIT; vs open: less blood loss, fewer complications. Morbidity plateaus at 18–20 robotic PN/year.
- **Positive margins (0–7%)** matter only in high-risk tumours (pT2–3, grade 3–4) → **surveillance, not reoperation**.
- **Port-site recurrence ~0.1%** — spillage/technique, not gas → impermeable bag.
- Local recurrence after RN ~2% → resection + systemic therapy; salvage after PN/ablation often ends in RN (planes lost).

6 · Other renal procedures

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first proves cyst causes pain; excise wall, biopsy, fulgurate base

Procedure	Indication	Key points
Laparoscopic renal biopsy	Failed percutaneous, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex
Nephropexy	Descent >5 cm / 2 vertebrae supine → erect, with pain, reduced flow or obstruction	Usually right, slim young women; 20% asymptomatic. Capsule to psoas/quadratus fascia
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Suture neck, obliterate cavity with fat
Nephrolysis	Chyluria, loin pain–haematuria	Lateralise by cystoscopy; hilar lymphatic ligation ± omental wrap
Ablation	≤3 cm, comorbid, solitary, hereditary RCC	Percutaneous preferred. Cryo: two freeze–thaw cycles, iceball 0.5–1 cm beyond tumour. RFA: 60–100°C (>105°C vaporises), 1 cm margin, no real-time imaging; more retreatment

7 · Complications & management

- P1 Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy — *vascular injury commonest (<1/1000, fatal 8–17%); venous bleed control; gas embolism (left side down, head down, aspirate); pneumothorax; thermal bowel injury, CT with oral contrast; rhabdomyolysis*
- Overall **~15%**; PN > RN > simple. Learning curve ~20 nephrectomies. Robotic PN halves open conversion. Age not a major risk; conversions cluster in infective disease.
- Vascular injury at **access** or hilum. Device failure: staplers > non-locking clips > locking clips (mostly operator error). **Left RN: SMA mistaken for renal artery** → bowel necrosis.

ALGORITHM

Major bleed: locate → temporary control (grasper/pressure) → inform anaesthetist (blood, open set) → clear field, raise pneumoperitoneum or add trocar → definitive clip/suture → else convert and call vascular surgeon

- Hilar injury causes: dividing before identification, sharp hilar dissection, stapler not covering vessel. Venous ooze → pressure; brisk arterial bleed → hand through lower midline incision. **Never clip or suture blindly.**
- End of case: **lower insufflation pressure** to unmask venous bleeding (bed, adrenal, mesentery, gonadal vessels, ureteric stump).
- Post-op haemorrhage → CT; arterial source → angioembolisation.
- **Haematuria ~day 12 after PN** → pseudoaneurysm/AVF → selective embolisation.
- **Urine leak** after PN (0.8–10%; large/hilar tumour, collecting-system repair, long WIT): percutaneous drain → stent + catheter or nephrostomy.
- **Bowel injury** (<1%, mortality ~3%; thermal ≈ half, 32% at access): **pain at a single trocar site**, distension, diarrhoea, **normal or low white count**. Stable → laparoscopic repair; unstable/gross spillage → laparotomy.
- **Solid organs:** spleen (left traction), pancreas — injury insidious, inspect at end; gallbladder → cholecystectomy.
- **Diaphragm**: sudden airway-pressure rise → purse-string, chest X-ray. Unrecognised pneumothorax ~0.3%.

- **Neuromuscular** (2.7%): flank rhabdomyolysis → hydration; axillary roll protects brachial plexus. VTE lower than open.

> **KEY POINT**
> MIS complications are the open ones in altered form and timing: the lethal ones — access vessel injury, thermal bowel injury — happen out of view and present late or atypically.

8 · Laparoscopic live donor nephrectomy †

- P4 P1 Laparoscopic donor nephrectomy — *left preferred (longer vein), donor keeps better kidney; gonadal–ureter packet; artery then vein; no Hem-o-lok on donor artery; mortality 3/10 000; lifetime ESRD risk*
- † Complications include chylous ascites; donors need lifelong annual BP and renal function checks.

143 Focal Therapy for Renal Tumours

1 · Rationale & indications

- Small RCC rarely kills; CKD and comorbidity do → minimise competing-risk harm *→ Ch 137*.
- **Partial nephrectomy (PN) = gold standard** for cT1a; thermal ablation (TA) – cryoablation (CA), radiofrequency (RFA), microwave (MWA) – the alternative.
- **Indications:** high surgical risk; solitary kidney; bilateral tumours; hereditary syndromes (VHL) needing repeat treatment. Faster recovery, fewer complications; renal function ≈ or better than PN.

KEY POINT
Trap: TA is never the default for a fit patient.

Guideline	Position on TA
AUA 2021	Consider for cT1a **≤3 cm**, esp. comorbid; PN prioritised; percutaneous preferred
EAU 2022	TA not routinely **>3 cm**; CA not **>4 cm**; for frail/comorbid
NCCN 2022	**Only one allowing cT1b** (non-surgical candidates); may need multiple treatments
ASCO 2017	Option if complete ablation achievable

2 · Selection & work-up

ALGORITHM
Counsel (age, comorbidity, CKD risk, size/location) → biopsy → re-counsel with histology → PN / surveillance / percutaneous ablation

- **AUA: biopsy all**, ideally at a **separate earlier session** (repeatable, fewer benign masses ablated). Routine post-ablation biopsy adds nothing; biopsy only a new enhancing lesion.
- Treat UTI first – colonised urine (ileal conduit) risks abscess.
- Complication predictors: RENAL score (low 4–6, moderate 7–9, high 10–12), ASA >3 (~3×), $(MC)^2$ (CA only). Day-case if ≤3 cm, low complexity, no anticoagulation.

3 · Mechanisms

- P2 Biological effects of cryosurgery — *freeze/thaw cell injury, vascular thrombosis, apoptosis, immunomodulation; argon cools/helium warms; kidney dies at −19.4 °C; <−20 °C only 3 mm inside iceball edge*
- Heat and cold: **central** necrosis + **peripheral sublethal** zone (apoptosis, ischaemia, immune). Cryo target **≤−40 °C**; iceball **5–10 mm beyond tumour** — every cryo rule pushes the −40 °C isotherm beyond the tumour edge.
- Heat: **≥60 °C** coagulative necrosis; ≥100 °C vaporisation/charring. Lethal: 45 °C × 60 min, 55 °C × 5 min, **70 °C × 1 min**.
- **Heat sink:** large vessels carry heat/cold away → perivascular survival. Counter with colder/longer freezes, more probes, or MWA.

4 · Choice of energy

RFA electrode	Feature	Strength
Single monopolar	Uninsulated tip	Tumours ≤2 cm only
12-tine umbrella	Tines act as antennas	Most spherical zone
9-tine starburst	Thermistors in tines	Best reproducibility
Cool-tip	Saline-cooled 17 G electrode	Largest zone

	Cryoablation	RFA	MWA
Physics	Argon cools (**Joule–Thomson**); helium warms	450–1200 kHz → ionic friction	915 MHz/2.45 GHz → **dielectric hysteresis**
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	**Double freeze–thaw** (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	**Yes**	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	**Least**
Zone visible	**Iceball**	**No** – impedance/temp	Gas bubbles on US
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

KEY POINT

Why RFA is self-limiting: charring raises impedance and halts current (cap 105 °C, electrodes cooled). MWA *radiates* energy, works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- Cryo: 10-min freeze is a compromise (5 under-treats, 15 fractures); second freeze enlarges necrosis.
- RFA impedance protocol: 40–80 W, +10 W/min to 130–200 W until impedance 200–500 Ω. MWA: usual cT1 settings 45–65 W for 5–15 min.

5 · Technique

- **Percutaneous = standard**; laparoscopic equally effective but more morbid – keep for prior scarring or failed hydrodisplacement. GA preferred (jet ventilation reduces renal motion).

ALGORITHM

IV antibiotics → prone/flank on CT gantry → US, CT or MRI-guided placement → non-contrast CT confirms position → ablate → contrast CT (completeness, haematoma)

- **Hydrodisplacement:** saline, or 5% dextrose (**D5W for RFA**, †saline conducts current); aim colon >1 cm. Caution in cardiac failure.
- **Pyeloperfusion** (central tumours): ureteral catheter irrigated at the **opposite temperature** – cold for RFA/MWA, warm for CA.

- **CA:** probes **1–2 cm apart**; remove only after full thaw (else fracture, bleeding). Bleeding rises with size and probe number (>4 cm).
- **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; overlapping ablations for larger lesions.
- **MWA:** antennas tangential, never into collecting system; 1 antenna ≤2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm.

6 · Follow-up, success & salvage

- **Success = complete loss of contrast enhancement** on CT/MRI; imaging margin replaces pathological margin.
- **Incomplete ablation:** enhancement on *first* scan → repeat (same-session imaging should be standard). **Local recurrence:** *later* enlarges/enhances → biopsy. **Enlargement after any modality = recurrence until proved otherwise.**
- Appearance: CA shrinks >50% in year 1; RFA minimal shrinkage, **fibrotic halo**; MWA low-attenuation zone that keeps contracting.
- **AUA:** CT/MRI ± contrast + chest imaging at **6, 12, 24, 36, 48, 60 months**; then shared decision. US not routine unless CEUS protocol.
- Salvage: re-ablation, PN, RN, or SBRT if comorbid.

7 · Oncological outcomes vs PN

- No RCT. LRFS worse after TA but **equal once re-ablation counted**; **MFS and CSS equal**; OS lower (older, comorbid).
- **Size predicts failure:** RFA worse ≥3 cm; T1b worse than T1a. No energy superior.

> **KEY POINT**
> **Why comparisons are biased:** benign and under-graded tumours flatter TA; occult pT3a penalises it; non-surgical candidates die of other causes first. EORTC 30904 (PN vs RN, ≤5 cm) had only 12 RCC deaths at 9.3 years – trials need huge cohorts.

8 · Complications & management

- Major (Clavien ≥3) **≤5%**, fewer than PN; percutaneous < laparoscopic; MWA lowest (network meta-analysis). Complications follow probe path and thermal field → prevention is planning.

Complication	Prevention	Management
Haemorrhage – commonest; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Start ablating (RFA/MWA coagulate); persistent → selective embolisation
Urinoma, fistula, PUJ/ureteric stricture	Stent + pyeloperfusion; CA for central; tangential antennas	Conservative or stent; large urinoma → percutaneous drain
Bowel/organ (tumour <1 cm away)	Hydrodisplacement, positioning	Colonephric fistula → stent + low-residue diet; persistent → diversion
Pneumothorax (above 12th rib)	Recognise during procedure	Aspirate; chest tube rarely
Nerve injury / RFA pad burn	Pads level on posterior thighs	Neuralgia self-limiting

Complication	Prevention	Management
Infection <1%, may be lethal; abscess at 1–6 months	Urine culture, prophylaxis	†Drain + antibiotics

9 · Newer modalities (investigational)

- **HIFU:** extracorporeal, thermal + cavitation above 3500 W/cm²; incomplete ablation (breathing, acoustic interference).
- **Histotripsy:** mechanical (non-thermal) focused ultrasound; no clinical data (HOPE4KIDNEY).
- **SBRT/SABR:** no anaesthesia; but no imaging endpoint, viable tumour on biopsy, lower CSS; possible salvage role.
- **IRE:** non-thermal membrane nanopores; spares matrix and vessels; poor oncological control.

144 Treatment of Advanced Renal Cell Carcinoma

1 · Overview

- P2 Metastatic RCC (mRCC): assessment and management, including recent concepts — *work-up, MSKCC/IMDC, first-line by risk, switch-mechanism later lines, CARMENA/SURTIME, metastasectomy 35–50% 5-yr, KEYNOTE-564, nccRCC*
- About **1/3** present with metastases; a further **20–40%** of localised tumours metastasise later. mRCC 5-yr survival ≈ **12%**; 10-yr <5%.
- Eras: cytokines → targeted therapy → **immune checkpoint (IO) combinations**. Adjuvant therapy *→ Ch 137*.

2 · Prognostic models

- **MSKCC (Motzer)**: cytokine era, 670 patients on IFN-α. **IMDC (Heng)**: first-line VEGF-targeted therapy; validated in 849.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< LLN	< LLN
Corrected calcium	**>10 mg/dL**	> ULN
LDH	**>1.5 × ULN**	—
Time factor	No prior nephrectomy	**<1 yr** diagnosis → systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median OS	20 mo	43.2 mo
Intermediate (1–2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

KEY POINT

Both score host reserve, tumour systemic effects and pace of disease; three factors shared. IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and short interval. The IMDC risk group **decides the first-line regimen**.

3 · Biology and drug targets

- **VHL** (3p) with elongins B/C and CUL2 tags hydroxylated **HIF-α** for degradation. VHL loss → HIF drives VEGF, PDGF, TGF-α, GLUT-1, erythropoietin. Drugs block ligand (bevacizumab), receptors (TKIs), HIF translation (mTOR) or HIF-2α (belzutifan); MET/AXL/FGFR escape explains broader later TKIs.
- **CTLA-4** and **PD-1/PD-L1** switch off T cells; inhibitors release the brake → autoimmune toxicity, benefit as OS and durability rather than PFS. VEGF inhibitors are immunomodulatory → rationale for IO + TKI.
- ccRCC chemoresistant (response 5.5–6%).

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A; IV	Bleeding, proteinuria, HTN
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, hand–foot syndrome (HFS); rarely used

Agent	Target / route	Signature toxicity / note
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off**	Diarrhoea, HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism**
Pazopanib	VEGFR, PDGFR, KIT	**Hepatotoxicity (↑ALT)**; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, **MET, AXL**	Fatigue, diarrhoea, HFS; frequent dose reduction
Lenvatinib	VEGFR1–3, **FGFR1–4**, PDGFR-α, RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; **IV weekly**	Mucositis, rash, ↑glucose, ↓phosphate, ↑cholesterol, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	**HIF-2α inhibitor**; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis, hypophysitis, vitiligo
Avelumab	**Anti-PD-L1**; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · Surgery in metastatic disease

- **CN rationale**: spontaneous regression <1%; bulky primary suppresses T cells; helps only as platform for systemic therapy. <1/4 reach systemic therapy if ECOG 2, multiple sites, brain/liver metastases.
- **CARMENA caveats**: ~45% poor risk; ~25% of CN arm off-protocol; stopped early, underpowered; sunitinib no longer standard — does **not** show harm in fit, low-burden, favourable patients.

ALGORITHM

mRCC → histology + IMDC risk + PS + metastatic burden

Fit, low burden, favourable features, resectable primary → consider upfront CN → systemic therapy

Intermediate/poor risk or high burden → systemic therapy first → deferred CN in responders

KEY POINT

CN helps only patients fit enough, and with disease slow enough, to reach effective systemic therapy; operating first on poor-risk patients mainly delays it.

- **Metastasectomy** favourable: complete resection, solitary, age <60, DFI >1 yr, lung, <4 cm, metachronous (markers of indolent biology).
- **Palliative**: nephrectomy for pain, haematuria, paraneoplastic syndromes — **angioembolisation for haematuria** first; only ~half of hypercalcaemic patients improve. Brain, weight-bearing, spinal metastases → surgery + radiation.

5 · Systemic therapy for ccRCC

	IFN-α	High-dose IL-2
Response	10–15%	15–20%
Durable complete response (CR)	<2%	**7–9%**; >60% of CRs stay disease-free
Regimen	Schedule/route immaterial	**600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	**Vascular leak**: hypotension, multiorgan failure; early mortality 2–5%
Current role	Not used first-line	Carefully selected fit patients

- IL-2: only agent approved on phase II response; low dose gives fewer durable CRs. Best candidates: clear cell, good PS, prior nephrectomy.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	**Active surveillance**, axitinib, high-dose IL-2
Intermediate / poor	Same three **+ ipilimumab + nivolumab**	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, **temsirolimus**

KEY POINT

Choosing: IO/TKI → higher response and PFS, better tolerated; IO/IO → deeper durable responses (CR 9%) with autoimmune toxicity; never compared directly. Ipilimumab + nivolumab restricted to intermediate/poor (favourable did better on sunitinib in CheckMate 214). Avelumab + axitinib not NCCN-preferred (no OS benefit). PD-L1 does not select. TKI or temsirolimus if IO contraindicated.

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	**Nivolumab or cabozantinib** (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, **belzutifan**, cabozantinib, lenvatinib + everolimus, **tivozanib**

- Tivozanib after ≥2 prior therapies. Failed: sunitinib + mTOR, nivolumab + sunitinib/pazopanib (toxic); bevacizumab + mTOR (RECORD-2, INTORACT).
- **Ipilimumab + nivolumab** toxicity: grade 3–4 **46%**; high-dose steroids 35%; discontinuation 22%; dose nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4 (CheckMate 016). Nivolumab alone vs everolimus grade 3–4: 19% vs 37%. Autoimmune events with ipilimumab predicted response.
- † Immune-related events: hold IO, steroids for grade ≥2 (prednisolone 1–2 mg/kg grade 3–4), taper ≥4 weeks; grade 4 → stop permanently.
- TKIs: dose reduction/interruption; avoid pazopanib with hepatotoxicity risk. mTOR: monitor glucose, lipids, phosphate, pneumonitis.

6 · Non-clear cell RCC

- 15–25%; no proven agent → **trial for all**. Papillary: sunitinib response 5–10%; temsirolimus beat IFN in nccRCC subgroup; sunitinib vs everolimus — trend to sunitinib. Chromophobe: trend to everolimus.
- **HPRC**: germline activating **MET** (chromosome 7) → type 1 papillary; sporadic papillary **trisomy 7 in >2/3**, somatic MET ~13%; foretinib/savolitinib/crizotinib active in MET-driven tumours.
- **HLRCC**: fumarate hydratase loss → type 2 papillary; glycolysis-dependent → **bevacizumab + erlotinib** (PR ~2/3).
- **Collecting duct → gemcitabine + carboplatin**. **Sarcomatoid**: gemcitabine-based; more PD-1-positive infiltrates.

KEY POINT
In nccRCC the syndrome names the driver, and the driver names the drug: VHL → VEGF/mTOR/HIF-2α; MET → MET inhibitors; FH → glucose restriction.

7 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs **18.4** mo, HR 0.89	No upfront CN in intermediate/poor
SURTIME	Immediate vs deferred CN	OS 15.0 vs **32.4** mo, HR 0.57	Systemic therapy first safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Later comparator
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; QoL/preference (PISCES) favoured pazopanib
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS ↑, OS not	Little used
Temsirolimus phase III	vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer	Lenvatinib + everolimus
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF	PFS HR 0.58; **OS HR 0.70**	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib

Trial	Design	Result	Take-home
TIVO-3	Tivozanib vs sorafenib, after 2–3 lines	PFS HR 0.73; OS same	Third-line tivozanib
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, **HR 0.73**; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS ↑; response 60 vs 40%	All risk groups
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS ↑; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, **HR 0.39**; response 71%	Largest PFS/response effect

★ Putting it together

A · Small renal mass

	AS	Ablation (Ch 143)	PN	RN
Best for	Frail, life expectancy <5 y; solid <2 cm or mainly cystic	cT1a **≤3 cm**, comorbid; biopsy first	cT1a standard; imperative if solitary, bilateral, hereditary, CKD	Only if high complexity + no CKD/proteinuria + new eGFR **>45**
Numbers	Growth 0.1–0.3 cm/y; metastasis <2%; treat if >3 cm or >5 mm/y	Cryo 2 freeze–thaw; RFA/MWA >60–70 °C	Trifecta: R0, no complication, WIT ≤25 min; margin width immaterial	New GFR ≈ GFR × contralateral split × 1.25
Weakness	Imaging adherence	More local recurrence (re-ablate); no pathology	Leak, bleeding, pseudoaneurysm	CKD 65% vs 20% after PN

B · Renal lesions

Lesion	Imaging	Genetics / IHC	Action
Simple cyst	≤20 HU, no enhancement	—	Nothing
Complex cyst	Bosniak v2019	—	≤2 cm surveil; >4 cm excise
AML	Fat ≤−20 HU; fat-poor = trap	TSC2/TSC1; HMB-45, Melan-A+	>6 cm, aneurysm >5 mm, bleed → embolise; TSC → everolimus
Oncocytoma	Central scar, enhancement inversion	KIT+, CK7 single cells; Birt–Hogg–Dubé	Can't exclude chromophobe → surveil/PN
ccRCC	Hypervascular/heterogeneous	3p/VHL → HIF-2α; CA-IX	PN/RN/TA/AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple losses; diffuse CK7	As RCC; good prognosis
Lymphoma/metastasis	Infiltrative, hypovascular	—	**Biopsy; don't operate**
UTUC	Filling defect, ~46 HU, cytology+	MSI → Lynch	Risk-stratify → KSM or RNU

C · UTUC: kidney-sparing (KSM) vs RNU

	KSM (URS, percutaneous, segmental)	RNU + bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low grade, no invasion on CTU; or imperative (solitary, CKD, bilateral)	ANY of multifocal, >2 cm, high grade, hydronephrosis, prior cystectomy, variant

	KSM (URS, percutaneous, segmental)	RNU + bladder cuff
Extras	URS each surveillance; UGN-101 mitomycin gel (low grade); ~3× re-interventions	≤12 wk; template LND; single post-op intravesical chemo
Systemic	—	Neoadjuvant platinum if post-op CrCl <50; adjuvant (POUT) ≥pT2/N+

D · Open vs minimally invasive

Open (Ch 141)	Laparoscopic / robotic (Ch 142)
Locally advanced (IVC thrombus needing bypass); cytoreduction; complex anatomy; prior surgery; trauma (midline, early vascular control)	Standard for organ-confined RN, most PN: less pain/blood loss/stay; same oncology; RP shorter

E · Metastatic RCC

ALGORITHM

1. IMDC (6 factors): KPS <80%, Hb < LLN, calcium/neutrophils/platelets > ULN, <1 yr to therapy → 0 favourable, 1–2 intermediate, ≥3 poor.

2. First line: IO + TKI (pembro + axitinib, nivo + cabozantinib, pembro + lenvatinib) any risk; ipi + nivo intermediate/poor.

3. CN: upfront only if fit, low burden, favourable; else deferred in responders (CARMENA, SURTIME).

4. Later lines: switch mechanism; belzutifan (HIF-2α) after IO + TKI.